

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
PART TEST – I
PAPER –1
TEST DATE: 19-11-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A, B, C, D

Sol. acceleration of the block will be :

$$\frac{F}{3m}, \text{ if } \frac{F}{\mu mg} < 6$$

$$\frac{F}{4m} + \frac{\mu g}{2}, \text{ if } 6 < \frac{F}{\mu mg} < 10 \text{ and } 3\mu g, \text{ if } \frac{F}{\mu mg} > 10$$

2. A, B, D

Sol. If $\frac{m}{M} \geq \frac{1}{2}$ the system is in equilibrium.

$$T + N = mg \quad \dots(i)$$

$$2T - 2N = Mg$$

$$T - N = \frac{Mg}{2} \quad \dots(ii)$$

Solving (i) and (ii)

$$mg \geq Mg/2$$

$$\Rightarrow M \leq 2m$$

3. A, B, D

$$\text{Sol. } mg - T = ma \quad \dots(i)$$

$$T - F = Ma_0 \quad \dots(ii)$$

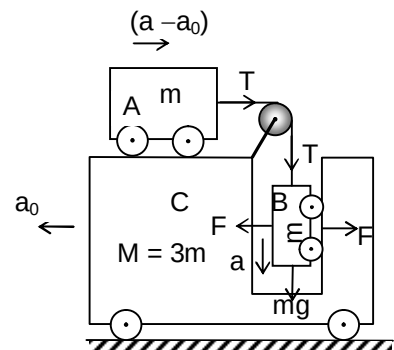
$$T = m(a - a_0) \quad \dots(iii)$$

$$F = ma_0 \quad \dots(iv)$$

Solving we get

$$a_0 = \frac{g}{9} \text{ and } a = \frac{5g}{9}$$

$$T = m(a - a_0) \Rightarrow T = \frac{4mg}{9}$$



4. B

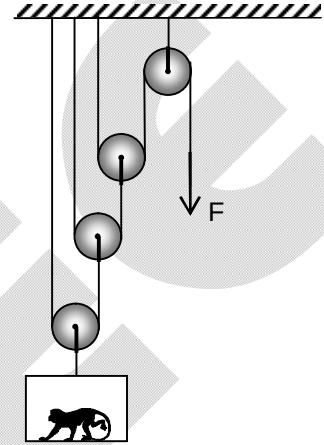
Sol. The acceleration tangent to the incline of the helix is given by $a_t = g \sin \theta$, where θ is the angle of the incline of the helix. As it accelerates, it moves faster, according to $v = a_t t$. Since this is motion in a circle, then there is a radial acceleration given by $a_r = \frac{v^2 \cos^2 \theta}{R} = \frac{(g^2 \sin^2 \theta \cos^2 \theta) t^2}{R}$. The magnitude of the acceleration is then: $a = \sqrt{a_t^2 + a_r^2}$

5. B

Sol. $m = 1 \text{ kg}$ and $M = 9 \text{ kg}$

$$2(2F - mg) - mg = \frac{(M + m)g}{2}$$

$$4F - 3mg = 5mg \Rightarrow F = 2mg = 20 \text{ N}$$



6. A

Sol. Let the horizontal component of the velocity of the projected ball be u_0 and the vertical component v_0 . The ball reaches the wall at point B after time, $t_0 = \frac{v_0}{g}$, $H = \frac{v_0^2}{2g}$

$$\text{Distance of A from the wall, } d = u_0 t_0 = \frac{u_0 v_0}{g}$$

The ball rebounds from the wall with a velocity eu_0 . The ball falls back to the ground level in time

$$t_1 = \frac{v_0}{g}$$

$$d_1 = eu_0 t_1 = e \frac{u_0 v_0}{g}$$

Similarly

$$d_2 = 2e^2 \frac{v_0 u_0}{g}, d_3 = 2e^3 \frac{v_0 u_0}{g}, \dots, d_n = 2e^n \frac{v_0 u_0}{g}$$

After infinite bounces, when its vertical velocity drops to zero, the ball arrives back at point A.

$$d_1 + d_2 + \dots = d$$

$$1 = e + 2e^2 + 2e^3 \dots \infty = e + \frac{2e^2}{1-e}$$

$$e = \sqrt{2} - 1$$

7. B

Sol. I is true because $\vec{F}_{12} \Delta t + \vec{F}_{21} \Delta t = 0$

II is true because by strong form they have same line of action and hence

$$\vec{r}_1 \times \vec{F}_{12} + \vec{r}_2 \times \vec{F}_{21} = 0 = (\vec{r}_1 - \vec{r}_2) \times \vec{F}_{12} = (\vec{r}_2 - \vec{r}_1) \times \vec{F}_{21} = \vec{0}$$

III is not true because when the total force is zero, the torque is reference-point independent

$$\sum \vec{r}_i \times \vec{F}_i = \vec{d} \times \sum \vec{F}_i + \sum \vec{r}_i \times \vec{F}_i = \sum (\vec{r}_i + \vec{d}) \times \vec{F}_i -$$

IV is not true because in general the two displacements of the points of action of the two forces are different and hence

$$\vec{r}_1 \cdot \vec{F}_{12} + \vec{r}_2 \cdot \vec{F}_{21} = (\vec{r}_1 - \vec{r}_2) \cdot \vec{F}_{12} = (\vec{r}_2 - \vec{r}_1) \cdot \vec{F}_{21} \neq 0 \text{ in general.}$$

8. A

Sol. Mark the radius of the pulleys from top to bottom as $r_1, r_2,$

$r_3, r_4, r_5,$ and r_6 . We know that $r_1 = 15$ and $r_6 = 25$.

If the axis of the moving pulleys are connected to each other. The mass rises with speed v , then the speed of each piece of rope is the value indicated in the figure. Velocity of the peripheral points of the pulleys relative to the axis of the pulley, moving from top to bottom $6v, 4v, 2v, v, 3v, 5v$. These peripheral speeds can be expressed using the speed and radius of the pulleys:

$$6v = 2\pi r_1 f, \quad 4v = 2\pi r_2 f, \quad 2v = 2\pi r_3 f$$

that is

$$r_1 : r_2 : r_3 = 3 : 2 : 1$$

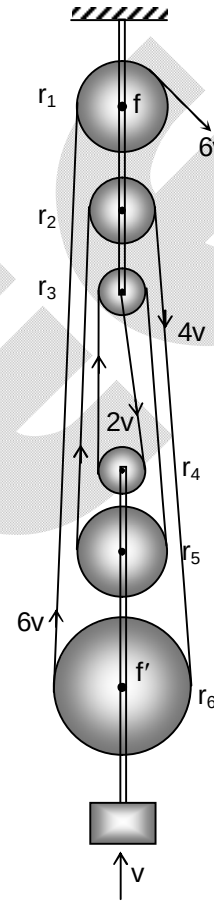
$$r_2 = 10 \text{ cm and } r_3 = 5 \text{ cm}$$

In a similar way, it can be assumed that moving pulleys

$$v = 2\pi r_4 f', \quad 3v = 2\pi r_5 f', \quad 5v = 2\pi r_6 f'$$

$$r_4 : r_5 : r_6 = 1 : 3 : 5,$$

$$r_4 = 5 \text{ cm and } r_5 = 15 \text{ cm}$$



9. A

Sol. Equations are:

$$f_1 - f_2 = Ma_c \quad \dots(i)$$

$$(f_1 + f_2)R = \frac{MR^2}{2} \alpha \quad \dots(ii)$$

$$2F - f_1 = Ma_1 \quad \dots(iii)$$

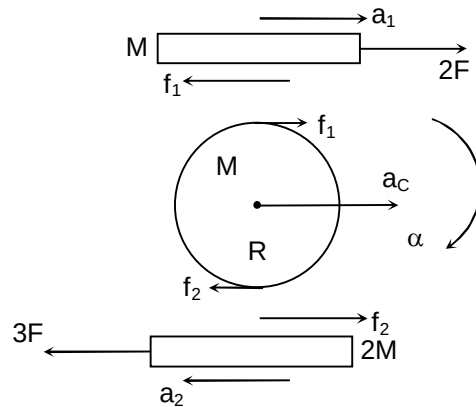
$$3F - f_2 = 2Ma_2 \quad \dots(iv)$$

Constraints are

$$\text{Lower point: } \alpha R - a_c = a_2 \quad \dots(v)$$

$$\text{Upper point: } \alpha R + a_c = a_1 \quad \dots(vi)$$

Solve for required quantities.



10. C

Sol. $t_0 = \frac{v}{2\mu g} = 1 \text{ sec}$

$$d = vt_0 - \frac{1}{2}\mu gt_0^2 = 1.5 \text{ m}$$

$$d_0 = \frac{1}{2}\mu gt_0^2 = 0.5 \text{ m}$$

11. D

Sol. $\frac{1}{2}\left(\frac{Mm}{m+M}\right)v^2 = mgR$

$$\Rightarrow \frac{mv^2}{4} = mgR$$

$$v = 2 \text{ m/s}$$

$$\mu mgL = mgR$$

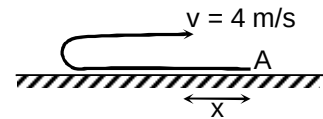
$$\Rightarrow \mu = R/L = 0.05$$

The displacement of P relative to the ground when Q reaches point-C will be $L' = \left(\frac{m}{m+M}\right)L$

SECTION - B

12. 3

Sol. Let at any instant, the end of the upper part is at a distance x from stationary end A. Then the centre of mass of the moving part is at a distance $x_{CM} = x + \left(\frac{\ell - x}{4}\right) = \frac{(\ell + 3x)}{4}$ from end A.



$$v_{CM} = 3 \text{ m/s}$$

13. 64

Sol. By energy conservation,

$$mgh = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 \quad \dots (i)$$

$$\frac{v_1}{v_2} = \frac{4}{3} \quad \dots (ii)$$

$$v_1 = \frac{4\sqrt{2}}{5} \text{ m/s}, \quad v_2 = \frac{3\sqrt{2}}{5} \text{ m/s}$$

$$\text{K.E.} = \frac{1}{2} \times 100 \times 10^{-3} \times \frac{16 \times 2}{25} = 64 \times 10^{-3} \text{ J}$$

$$n = 64$$

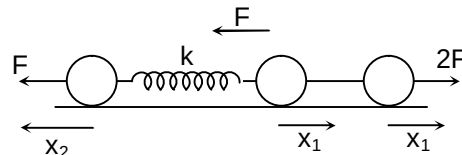
14. 2

Sol. According to work energy theorem

$$2Fx_1 - Fx_1 + Fx_2 - \frac{1}{2}k(x_1 + x_2)^2 = 0$$

$$\therefore x_1 + x_2 = x_{\max} = \frac{2F}{k}$$

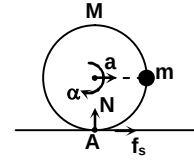
$$\therefore n = 2$$



15. 28

Sol. $r = \frac{mR}{m+M} = \frac{R}{3}$

$(M = 2m)$



$$I_A = \frac{3MR^2}{2} + 2mR^2 = 5mR^2$$

$$\tau_A = I_A \alpha$$

$$(M+m)g\left(\frac{R}{3}\right) = 5mR^2 \alpha$$

$$mgR = 5mR^2 \alpha$$

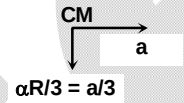
$$\alpha = g/5R$$

$$a = \alpha R = g/5 = 2 \text{ m/s}^2$$

$$(M+m)g - N = (M+m)\frac{a}{3}$$

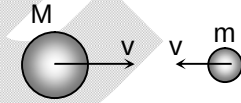
$$3mg - N = ma$$

$$N = 30 - 2 = 28 \text{ newton}$$



16. 3

Sol. Let the velocity of heavier ball after collision be v_1 and v'_1 relative to ground frame and centre of mass frame respectively.



$$v_{CM} = \left(\frac{M-m}{M+m}\right)v$$

Let the velocity of lighter ball relative to centre of mass frame be v'_2 after collision

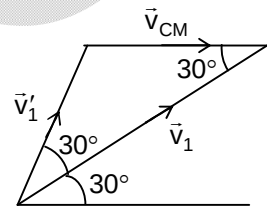
$$Mv'_1 = mv'_2 \Rightarrow v'_2 = \frac{M}{m}v'_1 \quad \dots(i)$$

Since, the collision is elastic

$$(v'_1 + v'_2) = 2v$$

$$\left(1 + \frac{M}{m}\right)v'_1 = 2v, \quad \left(\frac{M+m}{m}\right)\left(\frac{M-m}{M+m}\right)v = 2v$$

$$\frac{M}{m} = 3$$



$$v'_1 = v_{CM} = \left(\frac{M-m}{M+m}\right)v$$

17. 8

Sol. $v_1 = v_0 \sin 37^\circ = 10 \times \frac{3}{5} = 6 \text{ m/s}$

$$v_2 = v_0 \cos 37^\circ = 10 \times \frac{4}{5} = 8 \text{ m/s}$$

$$\int N dt = m(1+e)v_1 = \frac{3mv_1}{2} = \frac{3}{2} \times 4 \times 6 = 36 \text{ N-s}$$

$$J_{\max} = \mu \int N dt = 18 \text{ N-s}$$

Using conservation of angular momentum about point of contact P.

$$\frac{2}{5}mR^2\omega_0 + mv_2R = \frac{7}{5}mvR$$

$$(v = \omega R)$$

$$v = 10 \text{ m/s}$$

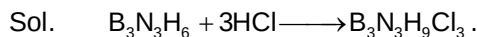
$$\text{Impulse due to friction on the ball, } J = m(v - v_2) = 4(10 - 8) = 8 \text{ N-s}$$

Chemistry

PART – II

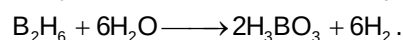
SECTION – A

18. C, D



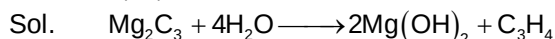
Boric acid behaves as a strong acid in presence of 1,2-diol.

B_2H_6 on hydrolysis produces H_3BO_3



Orthoboric acid is not a protonic acid.

19. A, B, D



20. B, C, D

Sol. Al_2Cl_6 has three centre four electron bond.

P – F axial bond length = 159.6 pm.

P – Cl equatorial bond length = 200.5 pm.

Bond angle $NH_3 = 107^\circ$.

Bond angle $NF_3 = 102^\circ$.

21. B

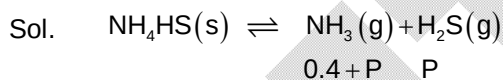
Sol. Number of local maxima = $n - \ell$

For 5d orbital, the number of local maxima = $5 - 2 = 3$

Number of angular node = ℓ

For 5d orbital, $\ell = 2$

22. D



$$2P + 0.4 = 0.64$$

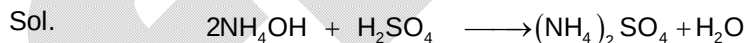
$$\therefore P = 0.12$$

$$K_p = P_{NH_3} \times P_{H_2S}$$

$$K_p = 0.52 \times 0.12 = 0.0624$$

$$K_p = 6.24 \times 10^{-2}$$

23. C



Initial 1 m.mol 0.4 m.mol 0

Final 0.2 m.mol 0 0.4 m.mol

The resulting solution is basic buffer

$$pOH = pK_b + \log \frac{0.8}{0.2}$$

$$= 4.76 + 0.6 = 5.36$$

$$pH = 14 - 5.36 = 8.64$$

24. C

Sol. Sn^{2+} is less stable than Sn^{4+} .

25. C

Sol. Sheet silicate - $(\text{Si}_2\text{O}_5)_n^{2n-}$

Pyrosilicate - $\text{Si}_2\text{O}_7^{6-}$

Orthosilicate - SiO_4^{4-}

Cyclic silicate - $(\text{SiO}_3)_n^{2n-}$

26. D

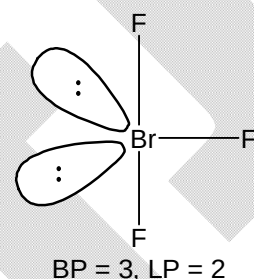
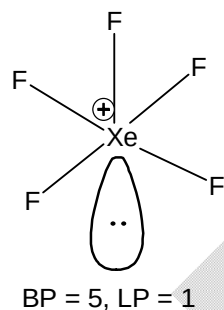
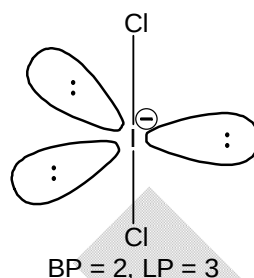
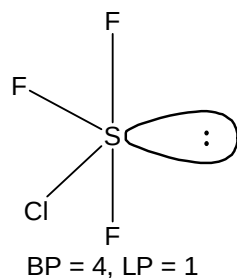
Sol. Conceptual

27. C

Sol. Conceptual

28. B

Sol.



SECTION - B

29. 17

Sol. $2\text{MnO}_4^- + 5\text{C}_2\text{O}_4^{2-} + 16\text{H}^+ \longrightarrow 2\text{Mn}^{2+} + 10\text{CO}_2 + 8\text{H}_2\text{O}$

$x = 2, y = 5, z = 16$

$a = 2, b = 10, c = 8$

$x + y + b = 2 + 5 + 10 = 17$

30. 14

Sol. Number of lone pair in Xe = 2

Number of lone pair in F = 3

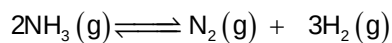
$\therefore$ Total number of lone pair of electron in $\text{XeF}_4 = 2 + (3 \times 4) = 14$.

31. 5

Sol. $V_n = \frac{2.18 \times 10^6 Z}{n} \text{ m/sec}, \quad n = \frac{2.18 \times 10^6 \times 1}{4.36 \times 10^5} = 5$

32. 7

Sol.



Initially (at 300K) 2.5 bar

$$\text{At 600 K} \quad 5(1-\alpha) \quad \frac{5\alpha}{2} \quad 3\left(\frac{5\alpha}{2}\right)$$

$$P_{\text{total}} = 5(1-\alpha) + \frac{5\alpha}{2} + 3\left(\frac{5\alpha}{2}\right)$$

$$= 5 + 5\alpha$$

$$= 5 \times 5(0.4) = 7 \text{ bar}$$

33. 4

Sol.

$$[\text{H}^+] = 10^{-4}$$

$$[\text{H}^+] = K_a \times \frac{[\text{acid}]}{[\text{salt}]}$$

$$10^{-4} = 2 \times 10^{-5} \times \frac{1 \times 0.2}{[\text{CH}_3\text{COONa}]}$$

$$[\text{CH}_3\text{COONa}] = 4 \times 10^{-2}$$

$$x = 4$$

34. 8

Sol.

$$\text{pH} = 8.56, \quad \text{pOH} = 5.44$$

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{NH}_4^+]}{[\text{NH}_4\text{OH}]}$$

$$5.44 = 4.74 + \log \frac{40 \times 0.2}{x \times 0.2}$$

$$0.7 = \log \frac{40}{x}$$

$$\log 5 = \log \frac{40}{x}$$

$$x = 8$$

Mathematics

PART – III

SECTION – A

35. B, C

Sol. $g_2(x) = \min_{0 \leq t \leq x} (g_1(t) + f(x-t)) = \min_{0 \leq t \leq x} (t^2 + (x-t)^2) = \min_{0 \leq t \leq x} 2 \left(\left(t - \frac{x}{2} \right)^2 + \frac{x^2}{4} \right) = \frac{x^2}{2}$

Similarly, $g_3(x) = \frac{x^2}{3} \dots ; g_n(x) = \frac{x^2}{n}$

$\Rightarrow \sum_{n=1}^{2024} g_n \left(\frac{1}{\sqrt{n+1}} \right) = \sum_{n=1}^{2024} \frac{1}{n} - \frac{1}{n+1} = \frac{2024}{2025}$ and $\sum_{n=2}^{2024} g_n \left(\frac{1}{\sqrt{n-1}} \right) = \sum_{n=2}^{2024} \frac{1}{n-1} - \frac{1}{n} = \frac{2023}{2024}$

36. A, B

Sol. Solving the two equation, we get $(x-y)(t(x+y) + t+1) = 0$ for $x=y$

$tx^2 + (t-1)x + \frac{1}{48} = 0$. If curve touch each other

$\therefore D = 0 \Rightarrow$ two rational values of t and for $y = -x - 1 - \frac{1}{t}$

$\Rightarrow tx^2 + (t+1)x + \frac{49}{48} + \frac{1}{t} = 0 \therefore D = 0 \Rightarrow$ Two irrational values of t

37. A, B, C

Sol. $2S_m = 2 \int_0^{m(m!) \pi} |\sin x| ([\sin x] + [-\sin x]) dx$

$S_m = -2m(m!)$

$R_n = \sum_{m=1}^n -2m(m!) = 2 - 2((n+1)!)$

38. C

Sol. $f(xf(y)) = x^3 y^a$, ($a \in \mathbb{R}$) put $x = 1$, we get $f(f(y)) = y^a$

Again put $f(y) = \frac{1}{x} \Rightarrow f(1) = \frac{1}{(f(y))^3} \cdot y^a$ and put $y = 1$, we get $(f(1))^4 = 1$

$\therefore f(1) = 1$, for $y = 1$, we have $f(x) = x^3$

and $\sum_{r=1}^n f(r) \cdot {}^n C_r = \sum_{r=1}^n r^3 \cdot {}^n C_r = \sum_{r=1}^n [r(r-1)(r-2) + 3r(r-1) + r] \cdot {}^n C_r = n^2(n+3)2^{n-3}$

39. A

Sol. Let $A(t_1, t_1^2)$ and $A(t_2, t_2^2)$ are the ends of the rod. Let $P(h, k)$ is midpoint $2h = t_1 + t_2$

$2k = t_1^2 + t_2^2$ also, $(t_1 - t_2)^2 + (t_1^2 - t_2^2)^2 = 16 \Rightarrow$ Curve traced $y = x^2 + \frac{4}{1+4x^2}$

Area = $2 \int_0^\infty \frac{4 dx}{1+4x^2} = 2\pi$

40. C

Sol.
$$I = \int \left(\frac{2 \cos x (\cos x + \sin x)}{1 + \sin 2x} - \frac{2x}{1 + \sin 2x} \right) dx = \int \frac{2 \cos x}{\sin x + \cos x} dx - \int \frac{2x}{(\sin x + \cos x)^2} dx$$

After using integration by parts in first part, $I = \frac{2x \cos x}{\sin x + \cos x} + c$

41. B

Sol.
$$I_2 = \int_{\sin^{-1} \frac{1}{e}}^{\frac{\pi}{2}} (1 + e \sin x + \ln \sin x)^{2023} \cos x dx, \text{ put } e \sin x = t$$

$$I_2 = \frac{1}{e} \int_1^e (x + \ln x)^{2023} dx \text{ and } I_1 = \int_1^e x \cdot \left(1 + \frac{1}{x}\right) (x + \ln x)^{2022} dx = \frac{e(e+1)^{2023} - 1}{2023} - \frac{e I_2}{2023}$$

$$\Rightarrow I_1 + \frac{e I_2}{2023} = \frac{e(e+1)^{2023} - 1}{2023}$$

42. D

Sol. (P) Equation of circle is $x^2 + \left(y - \frac{10}{3}\right)^2 = 1$

Consider a general point $(2t^2, 2t^3)$ on the curve $x^3 = 2y^2$, its distance from centre is

$$\ell^2 = 4t^4 + \left(2t^3 - \frac{10}{3}\right)^2 = 4 \left[t^4 + \left(t^3 - \frac{5}{3}\right)^2 \right]$$

$$\frac{d(\ell^2)}{dt} = 8t^2(t-1)(3t^2+3t+5) \Rightarrow t=1 \text{ is a point of local minima} \Rightarrow \ell_{\min} = \frac{2}{3}\sqrt{13}$$

(Q) Given curve is parabola of the form $(x-1)^2 = 4(y-2)$

Let AB is normal chord $A(2t_1+1, t_1^2+2)$, $B(2t_2+1, t_2^2+2)$

$$(AB)^2 = (t_1 - t_2)^2(4 + (t_1 + t_2)^2) \text{ also } t_2 = -t_1 - \frac{2}{t_1}$$

$$AB^2 = 16 \left(\frac{(1+t_1^2)^3}{t_1^4} \right); AB^2 \text{ is maximum at } t_1^2 = 2$$

(R) Let $t = -x - \frac{\pi}{6}$, $t \in \left[\frac{\pi}{6}, \frac{\pi}{4}\right] \Rightarrow \tan\left(x + \frac{2\pi}{3}\right) - \tan\left(x + \frac{\pi}{6}\right) + \cos\left(x + \frac{\pi}{6}\right)$

$$\Rightarrow \cot t + \tan t + \cos t \Rightarrow \frac{2}{\sin 2t} + \cos t \text{ of decreasing in } \left[\frac{\pi}{6}, \frac{\pi}{4}\right]$$

$$\Rightarrow \text{Greatest value occur at } t = \frac{\pi}{6} \Rightarrow \frac{11\sqrt{3}}{6}$$

(S) $(x-2)^2 + (y-2)^2 = 4$ (1)

Let equation of line is $y = mx$ put in equation (1)

$$(x-2)^2 + (mx-2)^2 = 4 \Rightarrow x^2(1+m^2) - 4x(1+m) + 4 = 0$$

$$\Rightarrow \text{Let M and N are } (x_1, y_1) \text{ and } (x_2, y_2) \Rightarrow (x_1 - x_2)^2 = (x_1 + x_2)^2 - 4x_1x_2 = \frac{32m}{(1+m^2)^2}$$

$$\text{Also, } y_1 - y_2 = m(x_2 - 24) \Rightarrow MN = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = (x_2 - x_1)\sqrt{1+m^2} = \sqrt{\frac{32m}{1+m^2}}$$

$$A = \text{Area of } \triangle MNQ = \frac{1}{2} \cdot MN \cdot (\text{length of perpendicular from Q on MN})$$

$$A^2 = \frac{32m}{(1+m^2)^2} \text{ and } A^2 \text{ is maximum at } m = \frac{1}{\sqrt{3}}$$

43. B

Sol. (P) $f(x) = \frac{1}{kx}$

(Q) $f(x) = 2x + 2$

(R) $f(x) = x^2 + xe^x$

(S) $(f(y) - f(x))y^y = x^x f\left(\frac{y^y}{x^x}\right)$ differentiate w.r.t x keeping y constant

$$-f'(x) \cdot y^y = x^x f\left(\frac{y^y}{x^x}\right)(1 + \ln x) - x^x f'\left(\frac{y^y}{x^x}\right) \frac{y^y (1 + \ln x)}{x^x}$$

$$f'(1) = 1, \text{ put } y^y = x^x, \text{ we get } -f'(x)x^x = x^x f'(1)(1 + \ln x) - x^x f'(1)(1 + \ln x)$$

$$f(1) = 0, \text{ hence } f'(x) = 1 + \ln x \Rightarrow f(x) = x \ln x$$

$$\Rightarrow \text{Area bounded between } y = x \ln x \text{ and } y = 2x - 2x^2 \text{ is } \frac{7}{12}$$

44. C

Sol. (P) $\frac{d(x^2 - y^2)}{(x^2 - y^2)^2} = 2d\left(\frac{x}{y}\right) \Rightarrow \frac{2x}{y} + \frac{1}{x^2 - y^2} = c$

(Q) $\int x^3 dx + \int \left(\frac{x}{y}\right)^2 d\left(\frac{x}{y}\right) = 0 \Rightarrow \frac{x^4}{4} + \frac{1}{3}\left(\frac{x}{y}\right)^3 = c$

(R) Given equation can be written as $2x^2y(x^2 dy + 2xy dx) + y^2 dy - x^2 dx = 0$
 $\Rightarrow 2x^2y d(x^2y) + y^2 dy - x^2 dx = 0$ integrating, $3(x^2y)^2 + y^3 - x^3 = c$

(S) $y dx + x dy + xy(y dx - x dy) = 0 \Rightarrow d(xy) + x^2y^2\left(\frac{dx}{x} - \frac{dy}{y}\right) = 0$

$$\Rightarrow \ln\left(\frac{x}{y}\right) - \frac{1}{xy} = c$$

45. A

Sol. (P) $f'(x) = 0 \Rightarrow 2x \sin \frac{\pi}{x} + x^2 \cos \frac{\pi}{x} \left(-\frac{\pi}{x^2}\right) = 0 \Rightarrow \tan \frac{\pi}{x} = \frac{\pi}{2x}$

$$\Rightarrow \text{Put } \frac{\pi}{x} = \theta \therefore \theta \in (\pi, 21\pi) \text{ as } x \in \left(\frac{1}{21}, 1\right)$$

$$\text{Now, number of solution of } \tan \theta = \frac{\theta}{2} \text{ in } (\pi, 21\pi) \text{ is } 20$$

(Q) Let a and b roots of $f(x) = 0$

$$\text{Now, } f(f(1)) = f(f(-2)) = 0, \text{ but } f(1) \neq f(-2) \Rightarrow f(1) = a \text{ and } f(-2) = b$$

$$\Rightarrow 1 + p + q + 3 = a, 4 - 2p + q + 3 = b$$

$$\Rightarrow \text{Adding above two equation } 2q - p + 11 = a + b \Rightarrow q = -\frac{11}{2}$$

$$\text{Also, } 3p - 3 = a - b \Rightarrow 9p^2 - 18p + 9 = p^2 - 4(q + 3) \Rightarrow 8p^2 - 18p - 1 = 0$$

$$\Rightarrow g(2) = 4 + 2(8p^2 - 18p) - q + 2 = \frac{27}{2}$$

$$(R) \quad f(x) = (x^9 - 9x^8) + (6x^8 - 54x^7) + (x^7 - 9x^6) + (x^6 - 9x^5) + (x^5 - 9x^4) + (x^4 - 9x^3) + (x^3 - 9x^2) + (x^2 - 9x) + x + 9$$

$$\Rightarrow f(9) = 18$$

$$(S) \quad L = e^{\lim_{x \rightarrow 0} \left(\tan^{-1} \frac{1}{x^2} + x^2 \sin \frac{1}{x^4} \right)} = e^{\frac{\pi}{2}}$$

SECTION - B

46. 450

$$\text{Sol. } I_1 = \lim_{x \rightarrow \infty} -x^2 \left(\left(1 + \frac{1}{x} + \frac{1}{x^3} \right)^{\frac{1}{3}} + \left(1 - \frac{1}{x} + \frac{1}{x^3} \right)^{\frac{1}{3}} - 2 \right) = \lim_{x \rightarrow \infty} -x^2 \left((1 + \alpha)^{\frac{1}{3}} + (1 - \beta)^{\frac{1}{3}} - 2 \right)$$

$$\left(\text{Where, } \alpha = \frac{1}{x} + \frac{1}{x^3}, \beta = \frac{1}{x} - \frac{1}{x^3} \right) = \lim_{x \rightarrow \infty} -x^2 \left(\frac{\alpha - \beta}{3} + \frac{1}{3} \left(\frac{1}{3} - 1 \right) \frac{\alpha^2 + \beta^2}{2!} + \dots \right)$$

$$= \lim_{x \rightarrow \infty} -x^2 \left(\frac{2}{3x^3} - \frac{1}{9} \left(\frac{2}{x^2} + \frac{2}{x^6} \right) + \dots \right) = \frac{2}{9}$$

$$I_2 = 2025$$

47. 2

Sol. Since $f(x)$ is monotonic $\Rightarrow f^{-1}(x)$ exist. Let $f^{-1}(x) = t \Rightarrow x = f(t)$

$$\int_{f(a)}^{f(b)} x(b - f^{-1}(x)) dx = \int_a^b f(t)(b - t) f'(t) dt = \frac{1}{2} \int_a^b (f^2(t) - f^2(a)) dt \Rightarrow k = 2$$

48. 56

$$\text{Sol. } I_2 = \int_0^1 \left(\frac{x}{7+x} \right)^{\frac{5}{2}} \left(\frac{1-x}{7+x} \right)^{\frac{7}{2}} \frac{dx}{(7+x)^2} \text{ put } \frac{x}{7+x} = t$$

$$I_2 = \frac{1}{9} \int_0^{\frac{1}{8}} (t)^{\frac{5}{2}} (1-8t)^{\frac{7}{2}} dt \text{ and now substitute } 8t = u \text{ after simplification } I_2 = \frac{I_1}{8^{\frac{7}{2}} \cdot 7^{\frac{7}{2}}}$$

49. 3

Sol. If $a < 0$, $x^2 + 2ax = 4x - 4a - 13 \Rightarrow x^2 + x(2a - 4) + 4a + 13 = 0 \Rightarrow D = 0$

$$\Rightarrow a = -1, 9 \text{ also } 0 < a + \frac{13}{4} < -2a \Rightarrow a \in \left(-\frac{13}{4}, -\frac{13}{12} \right) \Rightarrow a \in \left(-\frac{13}{4}, -\frac{13}{12} \right) \cup \{-1\}$$

For $a \geq 0$, $a = 9$, $x = -7$ and for $a = 9$, $x = -7$, $x^2 + 2ax < 0$

$\Rightarrow a \geq 0$ is not possible

50. 17

$$\text{Sol. } \lim_{x \rightarrow \infty} \frac{\int_0^x e^{f(t)} dt}{\frac{e^{f(x)}}{g(x)}}, \left(\frac{\infty}{\infty} \text{ form} \right) \text{ Applying L-Hospital Rule}$$

$$= \lim_{x \rightarrow \infty} \frac{(g(x))^2}{g(x) \cdot f'(x) - g'(x)} = \frac{(39)^2}{39 \times \frac{39}{17}} = 17$$

51. 4

Sol. $m = 0$, $M = \frac{\pi}{12}$, $a = 3^{\frac{1}{4}}$